Facial Pain Classification Using Deep Neural Networks for Clinical Decision Support

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Agenda

1. Overview/Motivation
2. Dataset/Preprocessing
3. Models & Methodology
4. Results & Conclusion

Motivation

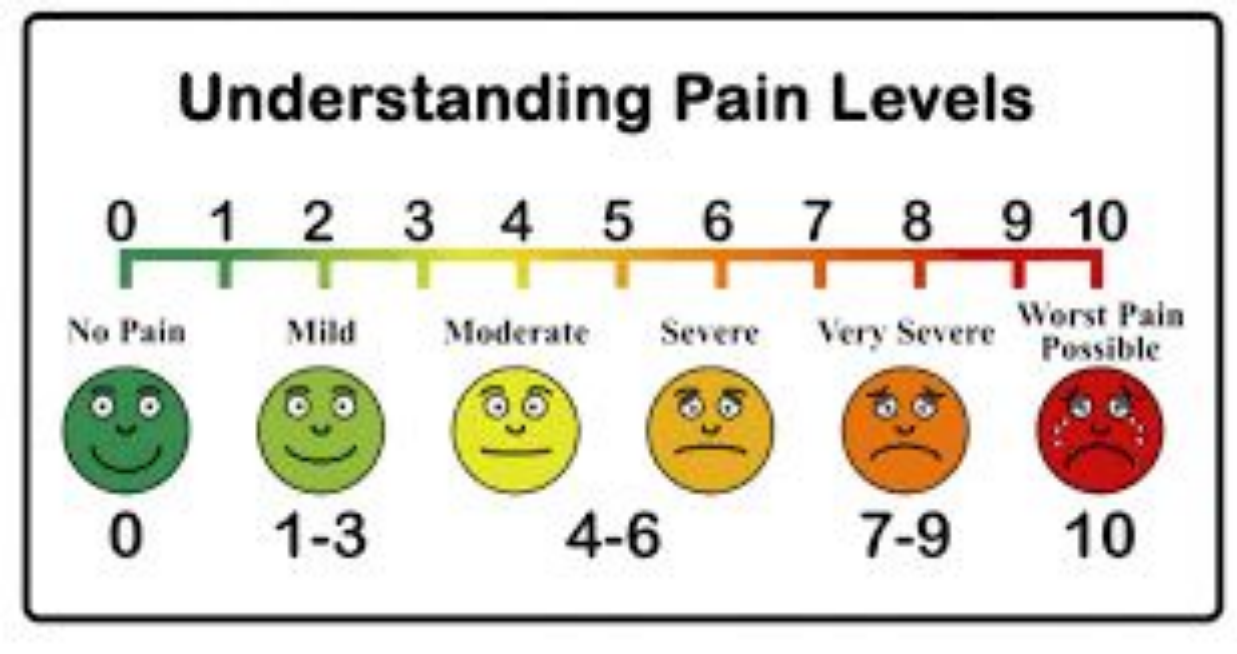
Problem

- Pain assessment is **highly subjective**
  - Relies on **self-report** or clinician **observation**
- Many patients **cannot communicate pain** effectively:
  - ICU patients
  - Post-surgical patients
  - Infants
  - Dementia / neurological conditions

Why this matters

- Misjudged pain can cause **under-treatment or over-treatment**
- Leads to:
  - Delayed recovery
  - Increased complications
  - Reduced quality of care

<u>Goal</u>: Build an **objective, automated system** that estimates pain intensity from facial images to **support clinical decision-making.**



Typical pain level reading chart in a healthcare facility: it is very **subjective and not always completely accurate**

Related Work

- Early methods used **facial action units (FACS)** to detect pain
- Deep learning (CNNs + transfer learning) now achieves much **better performance**
- Recent work uses attention and temporal models for improved accuracy

Our approach:

- Compare a **baseline CNN** with several transfer learning models (**VGG16, ResNet50, EfficientNet-B3**) to classify facial pain intensity
- Dual-input architecture that **combines full-face and mouth-region** features to capture both **global and localized** facial cues.

Dataset

- Based on AffectNet (real-world RGB images)
- Originally emotion-labeled, relabeled to pain levels:

Emotion	Pain Level
Neutral, Happy	No pain
Sad, Contempt	Mild
Fear	Moderate
Anger, Disgust, Surprise	Severe

- Because it is not clinically labeled, this serves as an **approximation of pain** for training

Preprocessing

- Resize images (fixed input size for NN)
- Normalize pixel values to stabilize training
- Face detection + cropping
- Data augmentation:
  - Flip
  - Rotation
  - Brightness
- Split:
  - Train / Validation / Test

Methodology

We compare:

- Baseline
  - Custom CNN
- Transfer Learning
  - VGG16
  - ResNet50
  - EfficientNet-B3
- Advanced
  - Dual-input model (face + mouth)

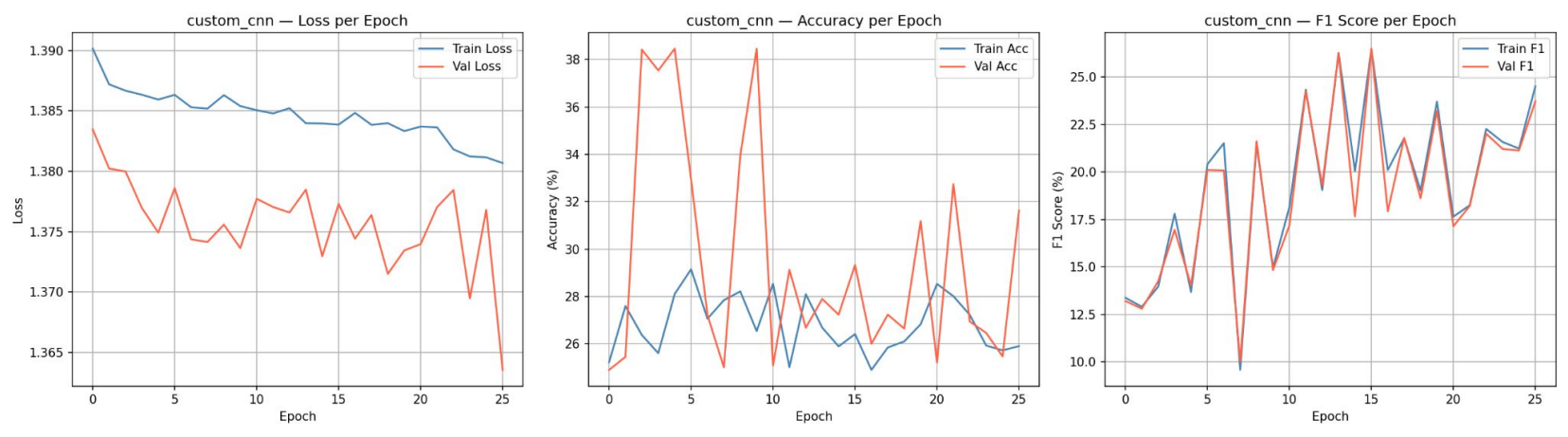
Training Strategy

- Loss: Weighted Cross-entropy
- Optimizer: Adam
- Batch size: ~32
- Epochs: ~50
- Techniques:
  - Early stopping
  - Dropout
  - Model checkpointing

```
criterion = nn.CrossEntropyLoss(weight=weights)
```

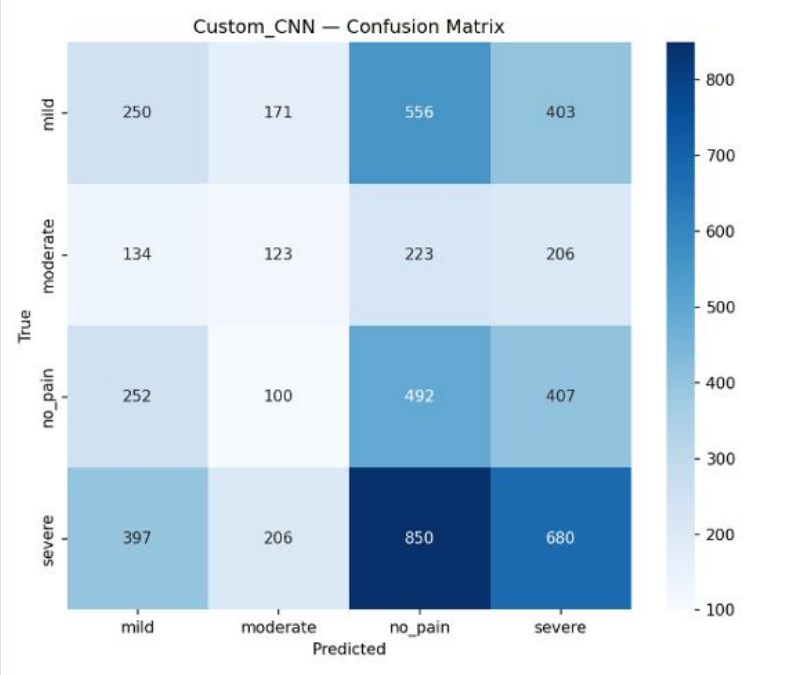
Custom CNN (Baseline)

- Trained from scratch
- Simple architecture
- **Low performance** (~26% F1-score)
- **Limited feature extraction**



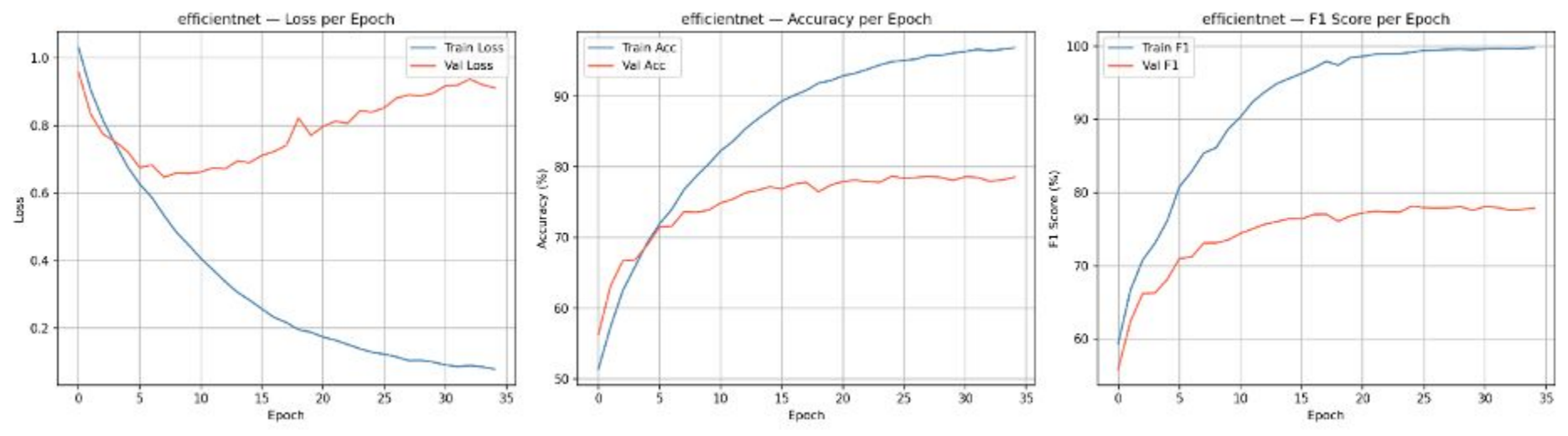
Custom CNN Confusion Matrix

- High misclassification across all classes
- Strong bias toward predicting no pain
- Poor separation between pain levels



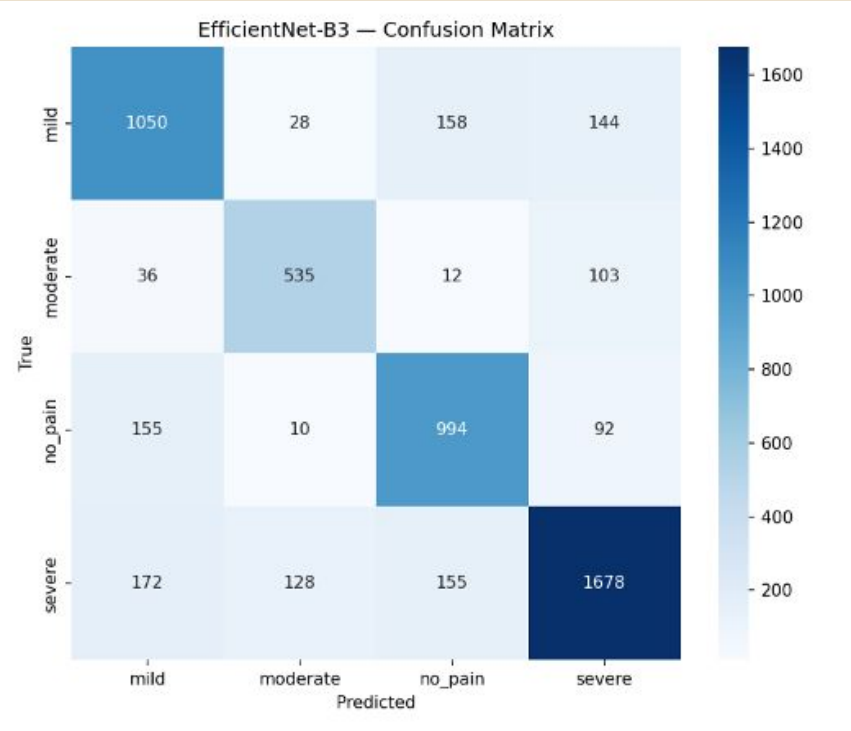
EfficientNet-B3

- Efficient scaling of model size
- Balanced performance (~78%)
- Slight **overfitting** observed



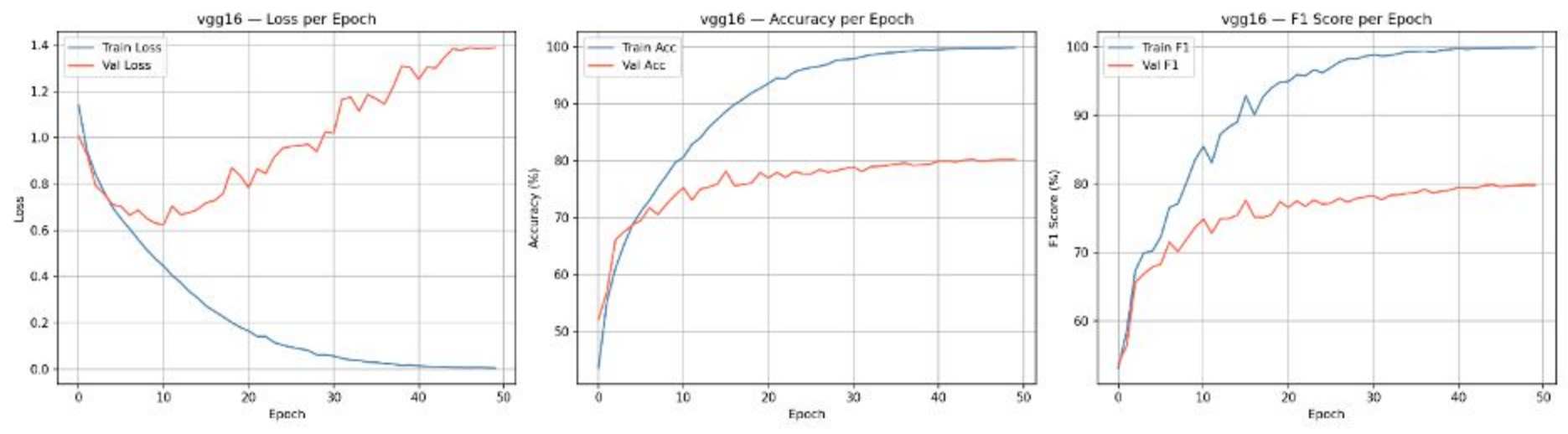
EfficientNet-B3 Confusion Matrix

- **Strong and balanced performance** across all classes with clear diagonal dominance
- **Minor confusion** persists between neighboring classes, but overall accuracy remains high and consistent



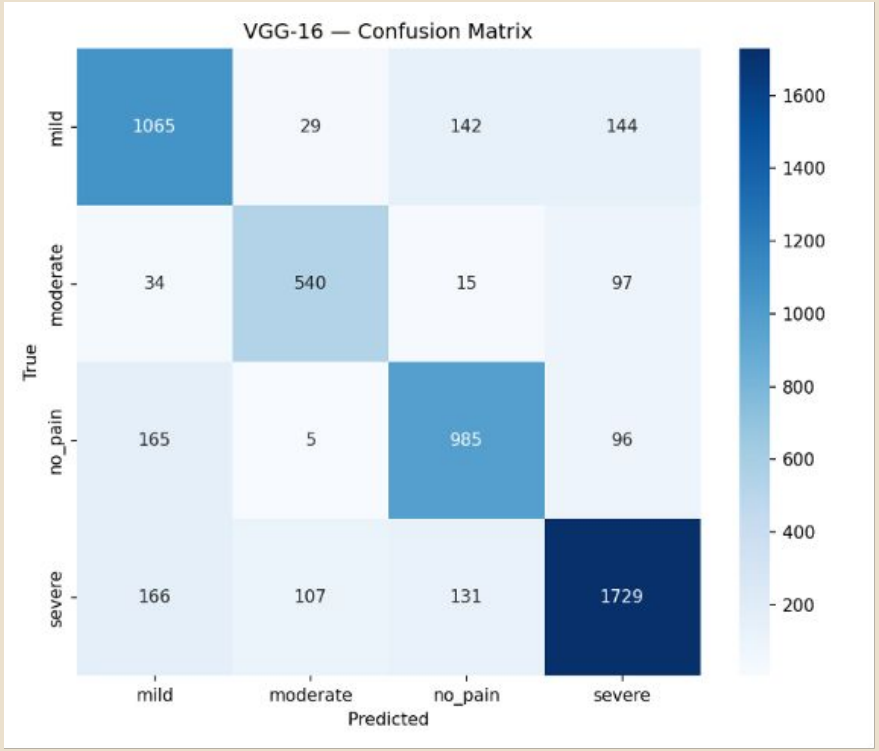
VGG 16

- Pre-trained deep network
- **Strong feature extraction**
- Best performance (~78.8% F1-score)
- Training accuracy improves steadily while validation performance plateaus (overfitting)



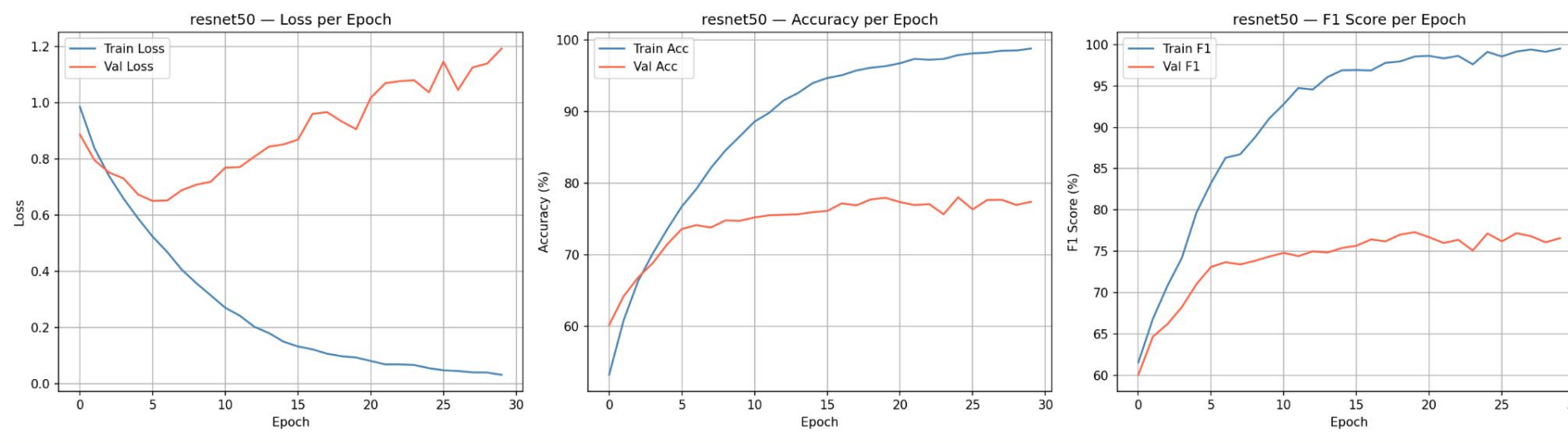
VGG 16 Confusion Matrix

- Strong diagonal dominance: **high accuracy across all classes**
- **Minor confusion** remains between adjacent classes, especially mild and moderate



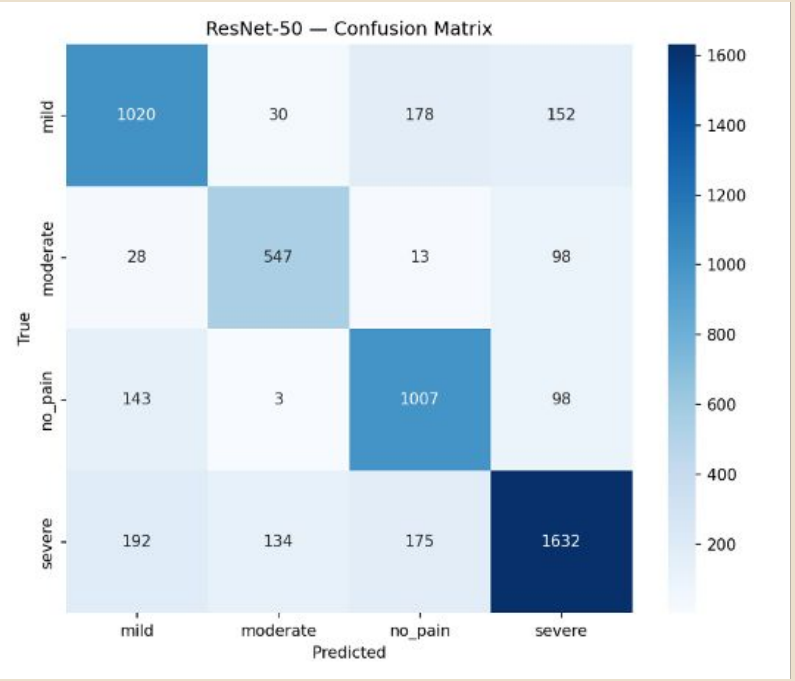
Resnet50

- Uses residual connections
- Enables **deeper learning**
- Strong performance (~77%)
- Mild **overfitting**



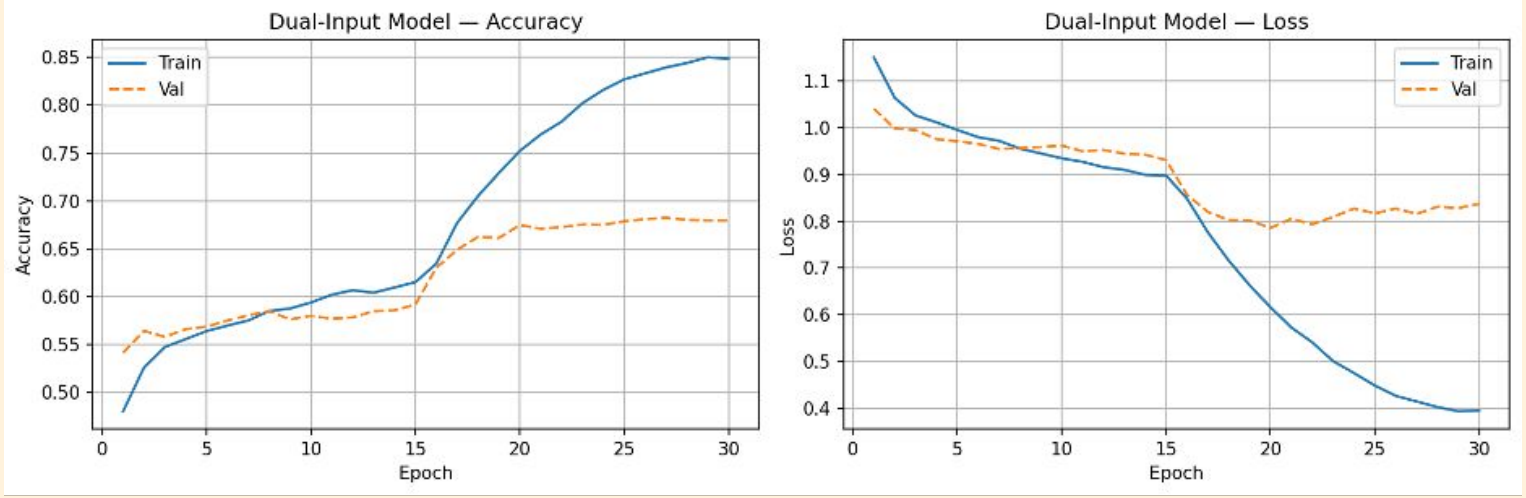
Resnet50 Confusion Matrix

- **Strong overall performance** with high correct predictions across all classes
- Slightly **more confusion than VGG16**, especially between mild and severe categories



Dual Input Model

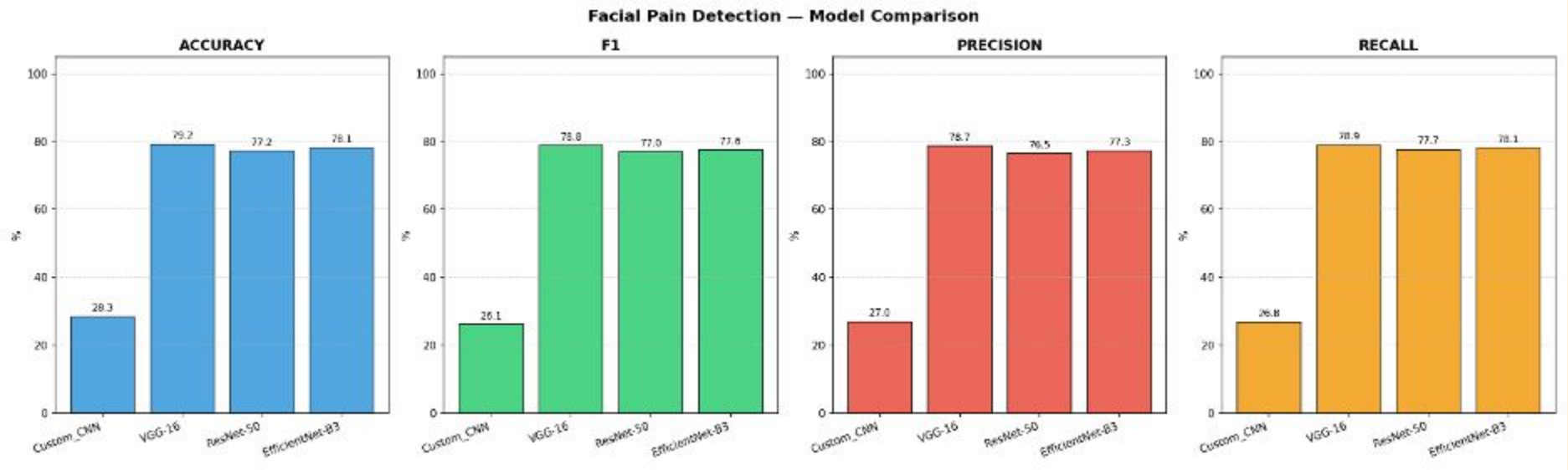
- **Two branches**- full face uses ResNet-50, mouth region uses ResNet-18
- Mouth branch captures key pain-related features (ex lip and chin tension)
- **Improves performance on subtle distinctions and enhances interpretability**



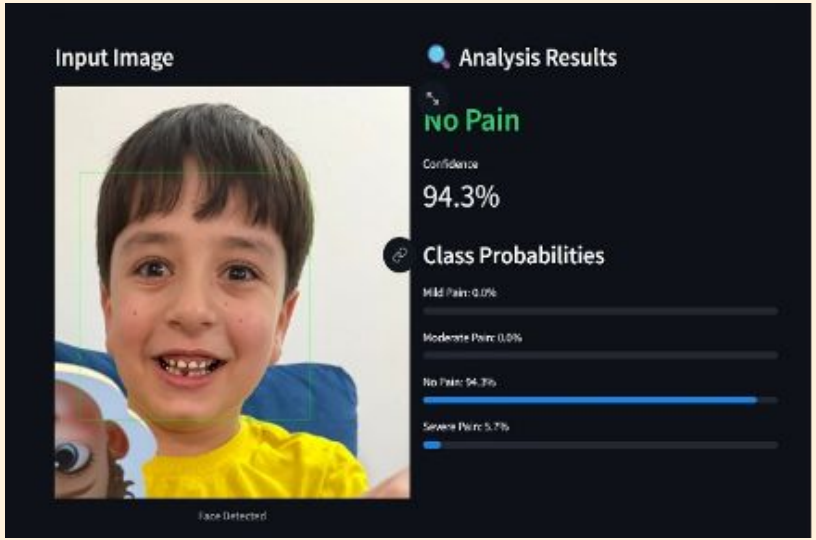
Results

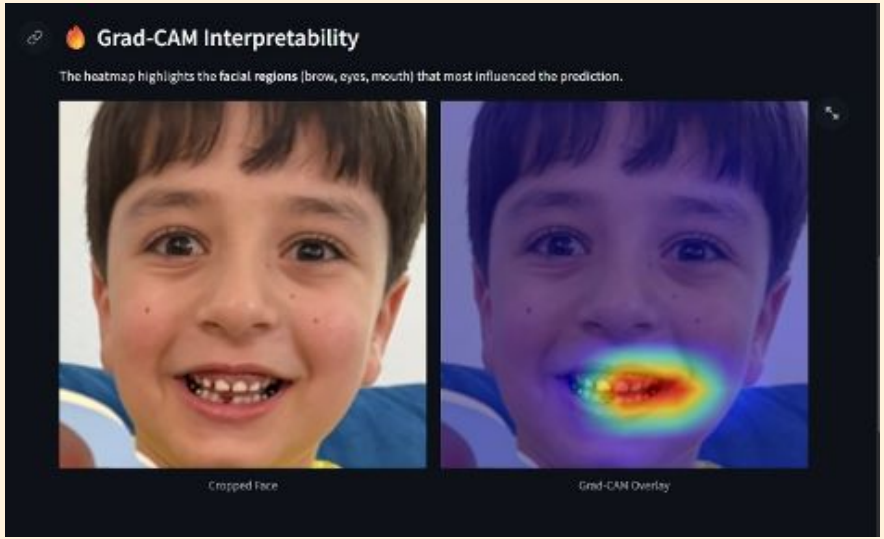
1 Mild differences between top models, but all outperform the baseline

2 Results confirm the importance of deep feature extraction for this task



Grad-CAM





Grad-CAM is a visualization technique that highlights the regions of an image that **most influence a model's prediction**. It helps us verify that the model focuses on **meaningful areas.**

Key takeaways

Transfer learning >> CNN

Pretrained models **outperform the CNN**, showing that **learned features** from large datasets are **crucial**

VGG 16 performs best

VGG16 achieves the best performance, indicating strong ability to **capture hierarchical facial features**

Mild/Moderate classes

These classes are **frequently confused** due to subtle and overlapping facial expressions

Mouth Region is critical

Key pain-related features are **concentrated around the mouth**, making it essential for accurate classification

Limitations

- **Not clinically labeled** dataset
- Emotion-based **approximation**
- **Static** images only
- Sensitive to pose and lighting

Future Work

- Use **clinical pain datasets**
- Incorporate video (temporal information)
- **Multimodal** learning (voice, signals)
- Improve robustness and fairness

Conclusions

- Deep learning **effective for pain classification**
- **Transfer learning** is essential
- Dual-input improves fine-grained detection
- Strong foundation for future work

Github Link
Demo

Ask us anything!